PHOTOREACTOR / WIRING GUIDE

Rev A hardware • SDL4-0001 controller + SDL4-0002 LED boards • 15 Oct 2026

COMPUTER
USB data → Pico

12 V DC ADAPTER
→ controller power input

CONTROLLER
SDL4-0001

REACTOR 1 → shown here
REACTOR 2–4: repeat

CABLE A • CONTROLLER → FIRST J1
10-pin keyed IDC, female ↔ female
Straight-through: pin n → pin n

FIRST LED BOARD
SDL4-0002 • LED A

J1 IN: 10-pin
J2 OUT: 6-pin JST PH
J3 FAN: 4-pin JST PH

STIRRING FAN
4 wires • 5 V / GND / tach / PWM

CABLE B • FIRST J2 → SECOND J1
Custom 6-pin JST PH → 10-pin IDC
Exact mapping below; not straight-through

SECOND LED BOARD
SDL4-0002 • LED B

J1 IN: 10-pin
J2 OUT: LEAVE EMPTY
J3 FAN: 4-pin JST PH

COOLING FAN
4 wires • 5 V / GND / tach / PWM

FAN LEADS
J3 → fan

PHR-4 plug
2.0 mm pitch
Terminate existing fan wires

CABLE A • 10-PIN PINOUT

1	LED A +	2	LED A −
3	LED B +	4	LED B −
5	Stir tach	6	Cool tach
7	Fan +5 V	8	Ground
9	Stir PWM	10	Cool PWM

All ten pins connect straight-through.
LED − is NOT ground.

CABLE B • CUSTOM HARNESS

First J2 OUT		Second J1 IN
1 LED B +	→	1
2 LED B −	→	2
3 Fan +5 V	→	7
4 Ground	→	8
5 Cool tach	→	5
6 Cool PWM	→	9

Leave second J1 pins 3, 4, 6, 10 unwired.

EACH FAN • J3 PINOUT
1 Ground
2 +5 V
3 Tach / RPM feedback
4 PWM speed control

FOUR-REACTOR TOTAL
1 controller • 8 LED boards • 8 fans
4 × Cable A • 4 × custom Cable B

Logical wiring diagram—not connector face views. Use numbered contacts; verify continuity before powering. Set controller JP_FANV to 2–3 (5 V).